

Shreya Jariwala

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RESEARCH PROFILE

I am a Geo-Data Scientist passionate about tackling pressing environmental challenges through AI. I am interested in deeply interconnected climate hazards, exploring and testing computational methods grounded in physical process reasoning to explain how climate change drives cryosphere change, hydrological extremes, and subsurface risk with the aim to contribute research that helps societies anticipate and adapt to a changing planet.

EDUCATION

Purdue University

West Lafayette, IN, USA

Master of Science in Geo Data Science (GPA: 4.00/4.00)

August 2022 – May 2024

- Research project: *Seismic Interpretation of Wind Turbine Noise Using Geophone Data, Illinois Basin*
- Outstanding GDSP Student Award (2023, 2024)

Indian Institute of Engineering Science and Technology

Shibpur, India

Master of Science in Applied Geology

August 2016 – May 2018

- Thesis: *Palaeoecological Analysis of the Infratrappean Horizon at Devrapalli Canal Section, Rajahmundry, Andhra Pradesh*

Shivaji Science College

Nagpur, India

Bachelor of Science in Physics, Chemistry & Geology

August 2013 – May 2015

EXPERIENCE

Independent Researcher – GeoAI Research Portfolio

May 2025 – Present

Self-directed

- * Conducted self-directed research applying physics informed, explainable machine learning to environmental hazards, spanning glacial hazard monitoring, InSAR based deformation analysis, flood forecasting, and geophysical inversion, using satellite remote sensing and deep learning methods.
- * Developed and designed the Earth Intelligence Platform, integrating eight Earth observation engines (location, satellite, terrain, weather, land-cover, risk, earth intelligence, etc) on Sentinel-2 and Copernicus DEM data into a single system, enabling a consolidated, decision ready view of environmental conditions for any given location.

GeoData Consultant

July 2024 – May 2025

Fugro – Geo-data Intelligence

Dubai, UAE

- * Designed automated Python workflows for seismic, borehole, and CPT interpretation, replacing manual processes across offshore wind energy site characterization projects.
- * Developed a CNN-based geophysical inversion model to estimate subsurface properties directly from survey data, automating traditionally manual interpretation.
- * Built interactive geo model visualization tools for geotechnical and geophysical data analysis, improving cross-disciplinary interpretation across project teams
- * Designed and architected structured geophysical database, enabling centralized data management across all interdisciplinary projects.

Assistant Manager – Geology

2018 – 2022

Hindalco Industries Limited

Jharkhand, India

- * Developed analytical dashboards to monitor the end-to-end mining production pipeline, enabling data driven decisions on ore quality and output
- * Led geological quality control through systematic ore characterization and spatial analysis, contributing to the division's best-quality bauxite performance across consecutive financial years.
- * Investigated mineral characteristics of residual bauxite reserves to identify quality-improvement opportunities, informing resource evaluation and reducing beneficiation costs.

SKILLS & CERTIFICATIONS

Programming: Python, SQL, R, MATLAB

ML & Data Science: TensorFlow, PyTorch, CNN, HDBSCAN, XGBoost, Random Forest, SHAP

Geospatial: Google Earth Engine, GeoPandas, Rasterio, xarray, QGIS, GDAL, OSM

Scientific Computing: HPC/GPU environments (Purdue Anvil, GPU clusters)

Certifications: Explainable AI (XAI) from Duke University, Fundamentals of Deep Learning from NVIDIA DLI)

RESEARCH PROJECTS

Independent Research Projects

- Trained a U-Net segmentation model on Sentinel-2 imagery to detect supraglacial lakes on Greenland's Russell Glacier, achieving a test IoU of 0.842 validated against a published reference dataset.
- Investigated climate drivers preceding the August 2026 Langtang Lirung collapse using ERA5-Land trend analysis and a convolutional VAE, finding the final week in the 99th percentile of 76 years atop a long-term warming trend.
- Benchmarked an LSTM rainfall runoff model against a conceptual hydrological model across UK catchments, using SHAP to show the model recovered each catchment's true hydrological response timescale directly from data.
- Built a hybrid physics-ML inversion pipeline combining Rayleigh-wave forward modeling with ML prediction of geotechnical parameters, uncovering a genuine bimodal non-uniqueness in the solution space.
- Built an independent MRV pipeline using GEDI and OCO-2/3 satellite data to verify registry carbon-credit claims for a Uganda REDD+ project, uncovering a 6.1% discrepancy against reported estimates.

Earth Intelligence System

- Developed and designed a modular geospatial intelligence platform integrating location intelligence, Earth observation, remote sensing, and spatial analytics for automated environmental monitoring and decision support.
- Designed scalable workflows for automated discovery, preprocessing, validation, and visualization of multi-source geospatial datasets.
- Built a Location Engine for automated Area of Interest generation, administrative boundary extraction, standardized spatial metadata, and coordinate validation using OpenStreetMap and GeoBoundaries.

Environmental Intelligence Projects

- Developed machine learning models to predict wildfire burned area using meteorological variables and Fire Weather Index components.
- Built a predictive analytics solution for Walmart Global Tech to identify drivers of PM_{2.5} concentration increases, supporting environmental monitoring and decision-making.

Vision Transformer for Avian Species Classification

- Designed, trained, and optimized a Vision Transformer (ViT) model on the NABirds dataset, achieving 96.13% classification accuracy.
- Applied transfer learning, data preprocessing, and model optimization techniques for fine-grained image classification.

AWARDS & RECOGNITION

- Outstanding Geo Data Science Student Award, Purdue University (2023 and 2024)
- PRIDE Award, Hindalco Industries Limited (2021) for outstanding contributions to mining operations and process improvement.
- STAR Graduate Engineer Trainee Award, Hindalco Industries Limited (2019) for excellence in geological characterization and beneficiation initiatives.
- Recognized for leading ore quality improvement initiatives that contributed to consecutive best-quality bauxite performance across the division.